

✖

Points:
0/2

1. A teacher logs roll numbers of students using a loop:

```
for i in range(1, 6):  
    print(f"Student Roll No: {i}")
```

Which statements are true?

(Select both the right answers)

☐ It prints from 0 to 5

☐ The loop executes 5 times ✓

☐ It runs infinitely

☒ It prints roll numbers from 1 to 5 ✓

✖

Points:
0/2

2. Neha's program reads a report file and catches errors if the file is missing:

```
try:  
    f = open("report.txt")  
    print(f.read())  
    f.close()  
except FileNotFoundError:  
    print("File is missing")
```

Which are true about this code? (Select both the right answers)

☐ It uses with open block

☒ It uses try-except for safe access ✓

☐ It avoids program crash if the file is missing ✓

☐ It writes data into the file

✔

Points:
1/4

3. Ravi is building a feedback system for a workshop and uses file modes in Python.

Match the correct pairs:

Create file but raise error if exists

"w"

✖

Correct answer: "x"

Add new feedback without deleting old

"w"

✖

Correct answer: "a"

Overwrite old summary with new one

"a"

✖

Correct answer: "w"

Read existing survey data

"r"

✔



Points:
1/1

4. A developer writes the following code to check login status:

```
username = "admin"
password = "admin123"

if username == "admin":
    if password == "admin123":
        print("Login Successful")
    else:
        print("Invalid Password")
else:
    print("Invalid Username")
```

What is the output for the given values?

Login Successful



Points:
0/1

5. An automation script checks if a folder exists before proceeding:

```
import os

if os.path.exists("myfolder"):
    print("Folder found.")
else:
    print("Folder missing.")
```

If the folder **doesn't exist**, what will be printed?

- ☒ None **x**
- ☐ Folder found.
- ☐ FileNotFoundError
- ☐ Folder missing. ✓



Points:
1/1

6. While displaying password hints, Tina wants to show the reverse of the user's password without revealing it directly.

```
password = "Secret123"
hint = password[::-1]
print(hint)
```

What will be printed?

- ☐ None
- ☐ Error
- ☒ **321terceS** ✓
- ☐ Secret123



Points:
1/1

7. Aman is collecting survey data and writing to a file. To avoid crashes on input error, he uses:

```
try:
    age = int(input("Enter age: "))
    with open("survey.txt", "a") as f:
        f.write(f"Age: {age}\n")
except ValueError:
    print("Invalid input! Please enter a number.")
```

What type of error is this code trying to prevent?

- ☐ NameError
- ☐ TypeError
- ☒ **ValueError** ✓
- ☐ FileNotFoundError



Points:
1/1

8. Arjun is maintaining scores of students in a quiz using a list:

```
scores = [85, 72, 90, 68, 94]
scores.sort(reverse=True)
print(scores[2])
```

Which score did the student ranked 3rd get?

- ☐ 68
- ☐ 90
- ☐ 72
- ☒ 85 ✓



Points:
2/2

9. An app gives users a warning if the temperature is very high:

```
temperature = 42
if temperature > 40:
    print("Heatwave alert!")
```

What other changes will still print the heatwave alert?

(Select all two right answers)

- ☒ temperature = 41 ✓
- ☒ Use if temperature >= 41: ✓
- ☐ if temperature < 40:
- ☐ if temperature == 38:



Points:
0/1

10. You're evaluating a condition in Python:

```
x = 5
y = 0
result = x and y
```

What will be the value of `result` ?

- ☐ True
- ☒ 5 ✗
- ☐ 0 ✓
- ☐ False



Points:
0/1

11. Tina wants to maintain a transaction log. Which mode will help her **add new entries without overwriting** old ones?

```
with open("log.txt", "___") as f:
    f.write("Transaction complete\n")
```

- ☐ "a" ✓
- ☒ "w" ✗
- ☐ "r"
- ☐ "x"



Points:
0/1

12. Ravi runs the following code to open a file and process it:

```
try:
    f = open("record.txt")
    data = f.read()
    f.close()
    print("File processed.")
except:
    print("An error occurred.")
```

What is the **issue with this code**, if any?

- ☐ The except block is syntactically invalid
- ☒ **finally block is missing, so file may stay open on error ✓**
- ☐ The file is always overwritten
- ☒ **File is not closed properly x**



Points:
4/4

13. Ritu is building an inventory tracker for a stationery shop. Each item is stored as:

```
item = {
    "name": "Marker",
    "quantity": 12,
    "price": 15.5,
    "in_stock": True
}
```

Which of the following are valid data types used in `item` ?

(Select all that apply)

- ☒ **float ✓**
- ☒ **int ✓**
- ☒ **bool ✓**
- ☒ **str ✓**



Points:
1/1

14. An education app calculates the grade based on marks:

```
marks = 82

if marks >= 90:
    grade = 'A'
elif marks >= 80:
    grade = 'B'
elif marks >= 70:
    grade = 'C'
else:
    grade = 'F'

print(grade)
```

What will be the student's grade?

✓



Points:
1/1

15. An administrator stores student roll numbers as a tuple to ensure they aren't accidentally modified:

```
roll_numbers = (101, 102, 103, 104)
roll_numbers[2] = 110
```

What will happen when this code runs?

- ☐ It removes the third roll number
- ☒ **It raises a TypeError ✓**
- ☐ It changes the third roll number to 110
- ☐ It executes without any issue



Points:
1/1

16. Ritu is organizing a 3-day workshop and is storing participant names for each day in a single list:

```
participants = ["Ankit", "Sneha", "Ravi", "Neha"]
participants.insert(2, "Priya")
print(participants)
```

What will be the output?

- ☒ **["Ankit", "Sneha", "Priya", "Ravi", "Neha"] ✓**
- ☐ ["Ankit", "Sneha", "Ravi", "Neha", "Priya"]
- ☐ ["Priya", "Ankit", "Sneha", "Ravi", "Neha"]
- ☐ ["Ankit", "Sneha", "Ravi", "Priya", "Neha"]



Points:
1/1

17. A sensor system checks values until a safe threshold is reached:

```
reading = 5
while reading < 10:
    print("Reading:", reading)
    reading += 2
```

How many times will the loop run?

✓



Points:
0/1

18. Anup is building a logout system for a web app. He wants to reset the user's access token using:

```
token = 17 # or any number
reset_token = token ^ token
print(reset_token)
```

What value will be assigned to `reset_token` after the operation?

✗

Correct answer: 0



Points:
0/1

19. Priya is developing a smart home control system. Two devices respond to these binary signals:

```
light = 10 # 1010
fan = 4 # 0100
signal = light ^ fan
print(signal)
```

What will be the final control signal output?

✗

Correct answer: 14

✗

Points:
0/1

20. The bank ATM system verifies balance before allowing a transaction:

```
balance = 5000
withdraw = 4500

if balance >= withdraw:
    if withdraw % 100 == 0:
        print("Transaction Approved")
    else:
        print("Amount must be in multiples of 100")
else:
    print("Insufficient Balance")
```

What will be printed?

Amount must be in multiples of 100 ✗

Correct answer: Transaction Approved

✗

Points:
0/1

21. A quiz app selects a random question from a list:

```
import random

questions = ["Q1", "Q2", "Q3", "Q4"]
print(random.choice(questions))
```

What will this print?

- ☒ Always "Q1" ✗
- ☐ Index of the selected question
- ☐ Any one from "Q1", "Q2", "Q3", "Q4" ✓
- ☐ Error due to wrong syntax

✗

Points:
0/1

22. You run the following Python code:

```
x = 5
y = 5.0
print(x == y, x is y)
```

What is the output?

- ☐ False True
- ☐ False False
- ☐ True False ✓
- ☒ True True ✗

✓

Points:
1/1

23. A fitness app checks if a person is a premium member **and** has an active subscription:

```
user_type = "Premium"
active = True

if user_type == "Premium":
    if active:
        print("Access granted")
```

Which combinations will still grant access?

- ☐ user_type == "Basic" and active == True
- ☐ user_type == "Premium" and active == False
- ☒ user_type == "Premium" and active == True ✓
- ☐ user_type == "Guest"

✗

Points:
0/1

24. Nikita is sending newsletters and wants to personalize the message by adding the user's name in uppercase:

```
name = "kiran"
message = f"Hello {name.upper()}, welcome to the team!"
print(message)
```

What will be printed?

- ☐ Hello KIRAN, welcome to the team! ✓
- ☐ Hello {name.upper()}, welcome to the team!
- ☐ Hello kiran, welcome to the team!
- ☒ HELLO KIRAN, welcome to the team! ✗

✓

Points:
1/1

25. An elevator panel encodes floor levels using a number. To reach a certain level, the control panel shifts the bit to the right:

```
floor = 8 # binary: 1000
print(floor >> 2)
```

Which floor number will it now represent?

2 ✓

✗

Points:
0/1

26. Meena is testing the logic of a card reader that combines two access flags:

```
flag1 = 5 # 0101
flag2 = 3 # 0011
combined = flag1 | flag2
print(combined)
```

What does the system output for the combined access level?

5 ✗

Correct answer: 7

✗

Points:
0/1

27. Nisha is writing student marks to a file:

```
with open("marks.txt", "w") as f:
    f.write("Ravi: 85\nNeha: 90")
print("File written successfully.")
```

After execution, what will happen?

- ☐ File is written and automatically closed ✓
- ☐ A FileNotFoundError is raised
- ☐ Raises a SyntaxError
- ☒ File remains open after execution ✗

✗

Points:
0/2

28. In a quiz game, the countdown starts at 5:

```
count = 5
while count > 0:
    print(count)
    count -= 1
```

Which of the following are true? (Select both the right answers)

- ☐ The loop starts at 0
- ☐ It stops when count == 0 ✓
- ☒ It prints 5 to 1 ✓
- ☐ The loop never stops



Points:
0/1

29. A teacher wants to log attendance for 5 students using a `for` loop:

```
for i in range(1, 6):  
    print(f"Student {i} present")
```

How many times will the message print?

6

Correct answer: 5



Points:
4/4

30. In an online student form, Neha encounters various errors while validating inputs and file access.

Match the correct pairs:

File not present

FileNotFoundError

int() used on a string like "abc"

ValueError

Name input expected but integer entered

TypeError

Dividing total by zero in average calc

ZeroDivisionError



Points:
1/1

31. Sanya is building a shopping cart. She wants to remove all instances of the product `"pen"` from the cart list:

```
cart = ["pen", "notebook", "pen", "pencil"]  
while "pen" in cart:  
    cart.remove("pen")  
print(cart)
```

What will be printed?

- ☐ ["pencil", "notebook"]
- ☐ ["pen", "pen", "notebook", "pencil"]
- ☐ ["notebook", "pen", "pencil"]
- ☒ ["notebook", "pencil"]



Points:
4/4

32. Nisha is building a budgeting app and logs each item in a dictionary.

```
entry = {  
    "item": "Pen",  
    "quantity": 2,  
    "cost": 12.5,  
    "in_stock": True  
}
```

Match the correct pairs:

12.5

Float ✓

2

Integer ✓

True

Boolean ✓

Pen

String ✓



Points:
1/1

33. Ravi is building a messaging app where certain offensive words must be replaced with *******. He uses the following logic:

```
msg = "This is a dumb idea"  
clean_msg = msg.replace("dumb", "****")  
print(clean_msg)
```

What is the output?

- ☐ This is a ***
- ☒ **This is a *** idea ✓**
- ☐ This is idea
- ☐ This is a dumb idea



Points:
2/2

34. Ravi is storing student results in a file:

```
with open("results.txt", "a") as file:  
    file.write("Suresh - Passed\n")
```

Which are correct statements? (Select both the right answers)

- ☐ It deletes the previous content
- ☐ It opens the file in read mode
- ☒ **The file remains safe (previous data stays) ✓**
- ☒ **The data is appended at the end ✓**



Points:
0/1

35. A school arranges 2 rows with 3 seats each. The layout is printed using nested **for** loops:

```
for row in range(1, 3):  
    for seat in range(1, 4):  
        print(f"Row {row} Seat {seat}")
```

What will be the total number of seat labels printed?

3 ✗

Correct answer: 6



Points:
0/1

36. Vikram is creating a school timetable using a nested list:

```
timetable = [  
    ["Math", "English", "Science"],  
    ["History", "Math", "PE"]  
]  
print(timetable[1][2])
```

Which subject is scheduled in the 3rd period of Day 2?

- ☐ PE ✓
- ☐ History
- ☒ Science ✗
- ☐ Math



Points:
0/1

37. An event planner uses a dictionary where each **event date (day, month)** is the key, and the value is the event name:

```
events = {  
    (15, 'Aug'): 'Independence Day',  
    (26, 'Jan'): 'Republic Day'  
}  
print(events[(15, 'Aug')])
```

What will be printed?

- ☐ Republic Day
- ☐ Independence Day ✓
- ☒ 15, Aug ✗
- ☐ KeyError



Points:
1/1

38. A customer is eligible for discounts based on the purchase amount:

```
amount = 1200  
if amount > 1000:  
    discount = 0.20  
elif amount > 500:  
    discount = 0.10  
else:  
    discount = 0  
print(discount)
```

What discount percentage will be applied?

✓



Points:
1/3

39. Ajay is designing a quiz app and wants to select questions and shuffle choices randomly.

Match the correct pairs:

Shuffle multiple answer choices

✗

Correct answer: shuffle(list)

Select one question from a list randomly

✗

Correct answer: choice(list)

Generate a 6-digit OTP

✓

✖

Points:
0/1

40. Manoj wants to read a file that may or may not exist:

try:

```
with open("data.csv", "r") as file:
```

```
    content = file.read()
```

except FileNotFoundError:

```
    print("File not found.")
```

If `data.csv` is missing, what will be printed?

- ☐ The program crashes
 - ☒ **File not found. ✖**
 - ☐ Nothing is printed
 - ☐ Error
-